LOGO

Offshore Wind Turbine Project – Report I

Part I.A: Wave dynamics – mean wave conditions and seasonal variability
at a candidate floating wind site off south-west Brittany

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Abstract

A 30-year (1996–2025) 3-hourly wave hindcast from the Copernicus Marine Service global wave reanalysis is analysed at a candidate offshore wind site located ≈50 km west of the Pointe du Raz (Brittany, France), in 68.5 m of water. The mean sea state is $\overline{H_{m0}} = 2.42$ m, $\overline{T_{m02}} = 6.48$ s, coming from 284° (WNW). No statistically significant trend is found in wave height, period or direction over the 30 years; interannual variability (e.g. the 2013/14 winter) dominates. The most frequent sea state is $H_{m0} \approx 1.6$ m, $T_{m02} \approx 4.75$ s. Waves come from the W–WNW sector 82 % of the time and are deep-water waves 95 % of the time (based on T_{m02}); most are weakly non-linear. A strong seasonal cycle is observed: winter H_{m0} is about twice the summer value.

1 Introduction and site description

The aim of the project is to design a floating offshore wind turbine (OWT) and its anchoring system. This report covers Part I.A: it characterises the mean wave conditions at the selected site and their seasonal and interannual variability, answering the three questions of the assignment (Sections 3.1–3.3). The extreme wave conditions (Part I.B) will be treated separately. These results will be used as boundary conditions for the wave propagation modelling (Part II) and the wave–structure interaction analysis (Part III).

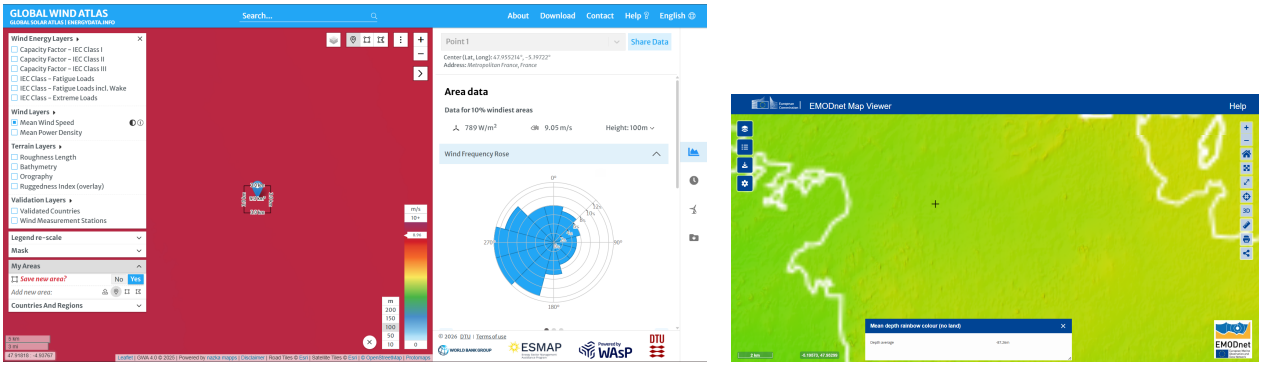
Site selection. The site is located in the Atlantic approaches to Brittany, south-west of the Iroise Sea (Table 1). It was chosen using the criteria introduced in class:

- **Wind resource.** The Global Wind Atlas gives a mean wind speed of 9.05 m/s at 100 m hub height (power density 789 W/m^2; Fig. 1a). This is above the 8.5 m/s "safe investment" threshold. The wind rose is dominated by the SW–W sectors.
- **Water depth.** The EMODnet bathymetry gives a mean depth of 68.5 m at the site. This is above the ≈60 m depth beyond which floating foundations become more economical than bottom-fixed ones [2].
- **Distance to shore.** About 50 km from Île de Sein and the Pointe du Raz. This is close enough for grid connection and far enough to limit visual impact and conflicts with coastal activities.

The seabed in this area deepens gently towards the west. Nearby points span roughly 68 m to 87 m (Fig. 1b), so the exact depth of the turbine location should be confirmed with a local bathymetric survey.

Table 1: Site characteristics and data sources.

Site coordinates	47.969 °N, 5.494 °W
Nearest wave-model grid point	48.0 °N, 5.4 °W (≈8 km from the site)
Mean water depth h (EMODnet)	68.5 m
Mean wind speed at 100 m (Global Wind Atlas)	9.05 m/s
Distance to the coast	≈50 km (Île de Sein / Pointe du Raz)
Wave data	CMEMS global wave reanalysis (MFWAM, 0.2°, 3-hourly) [1]
Period	01/01/1996 – 31/12/2025 (30 years, 87 664 sea states, no gaps)



(a) Global Wind Atlas: mean wind speed and wind rose at 100 m. **(b)** EMODnet bathymetry viewer near the site.

Figure 1: Site selection data.

2 Data and methods

Wave data. I used the Copernicus Marine global wave reanalysis (Météo-France MFWAM spectral model forced by ERA5 winds, 0.2° resolution) [1]. A 3 × 2 grid-point box around the site was downloaded with three variables: the spectral significant wave height H_{m0} (`VHM0`), the mean period T_{m02} (`VTM02`) and the mean direction the waves come from, θ_m (`VMDR`). All analyses use the single grid point closest to the site (48.0 °N, 5.4 °W). Across the six grid points of the box, the mean H_{m0} varies by less than 0.2 m, so the results are not sensitive to this choice. No buoy observations were available at this site to validate the model.

Mean and seasonal analysis. I computed the averages of H_{m0} and T_{m02} over the whole record. Directions are angles, so their average is the circular mean $\bar{\theta} = \operatorname{atan2}(\overline{\sin\theta}, \overline{\cos\theta})$. The circular spreading is $\sigma_\theta = \sqrt{-2\ln R}$, where R is the mean resultant length. To look for trends, I computed annual means and tested them with a least-squares fit and with a non-parametric Mann–Kendall test (Theil–Sen slope). The monthly climatology is the mean over all years for each calendar month.

Wave regimes. The deep-water wavelength is $L_0 = gT^2/2\pi$. Following linear wave theory, a wave is in deep water when $h > L_0/2$ (i.e. $T < \sqrt{4\pi h/g}$) and in shallow water when $h < L_0/20$. Wave non-linearity is measured by the steepness H_{m0}/L_0:

- Waves are treated as *linear* when $H_{m0}/L_0 < 0.02$. Below this value, the second-order Stokes correction to the crest height, $ka/2 = \pi H/(2L)$, is below 3 %.
- Waves become *weakly non-linear (Stokes)* as the steepness increases.
- Waves reach the breaking limit at $H/L \approx 1/7$.

The Ursell number $U_r = HL^2/h^3$ is used to check whether the cnoidal (shallow-water) regime is relevant.

All figures and numbers in this report come from one script (`analysis.py`, provided with the report).

3 Results

3.1 Question 1 – Mean wave conditions and long-term trends

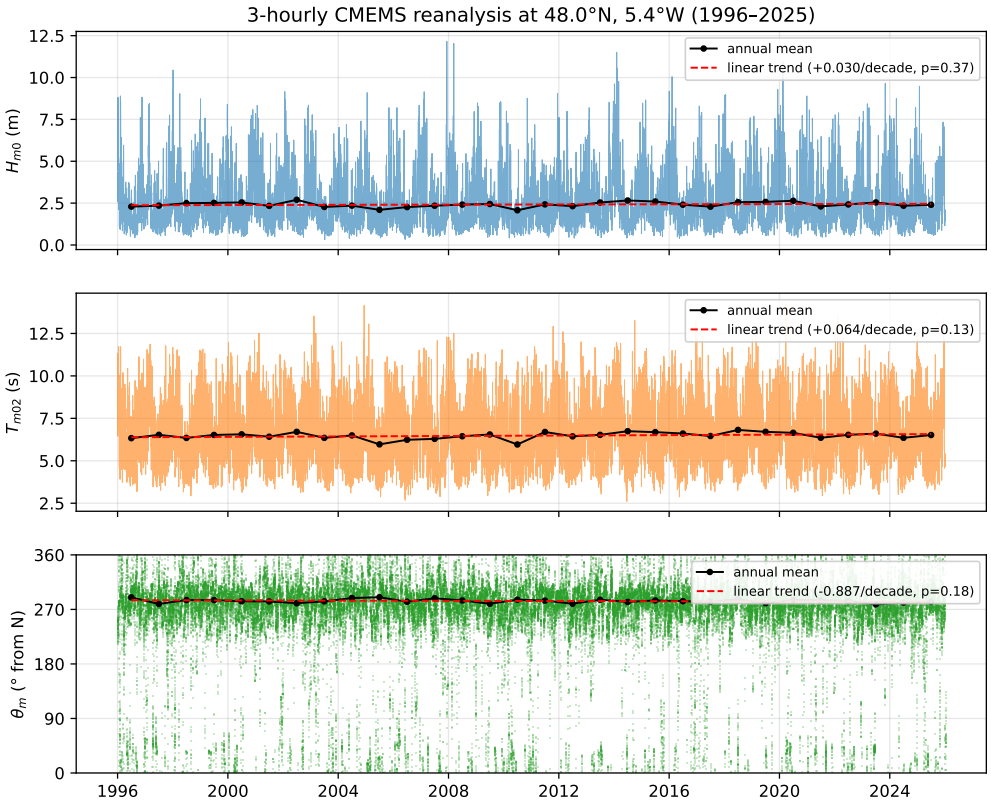


Figure 2: Time series of H_{m0}, T_{m02} and θ_m (3-hourly), with annual means (black) and linear trends of the annual means (red, slope and p-value in the legend).

Table 2: Long-term statistics (1996–2025) and trends of the annual means. Trends are given per decade; p-values are from the least-squares fit and the Mann–Kendall test.

	Mean	Std / spread	Range of annual means	Trend /decade	p (LS / MK)
H_{m0} (m)	2.42	1.40	2.07 (2010) – 2.70 (2002)	+0.03	0.37 / 0.34
H_{m0} annual 99th pct. (m)	6.92	–	5.92 (2010) – 8.70 (2014)	+0.02	0.85 / 0.86
T_{m02} (s)	6.48	1.67	5.96 (2010) – 6.82 (2018)	+0.06	0.13 / 0.13
θ_m (° from N)	283.8	38.3 (circ.)	278.4 (2023) – 290.1 (2005)	−0.9	0.18 / 0.21

The mean sea state at the site is $\overline{H_{m0}} = 2.42\,\text{m}$ (median 2.04 m), $\overline{T_{m02}} = 6.48\,\text{s}$ and $\bar{\theta}_m = 283.8°$, i.e. waves come from the WNW (Table 2). The largest sea state in the record is $H_{m0} = 12.16\,\text{m}$, on 9 December 2007. The time series (Fig. 2) is dominated by the annual cycle, with large winter storm peaks above 8 m almost every year. Directions are concentrated between 250° to 320°. Rare easterly events (4.5 % of the time) correspond to offshore winds blowing from land.

Q: *Do any trends in the wave height, period, or direction exist over the time period? Do you expect there to be changes in the mean conditions during the 30-year lifetime of the wind turbine? If so, why?*

No statistically significant trend is detected. The fitted slopes are very small: +0.03 m/decade for H_{m0} (1 % of the mean per decade), +0.06 s/decade for T_{m02} and −0.9°/decade for the direction. All p-values are larger than 0.1 with both tests, and the storm statistic (annual 99th percentile of H_{m0}) shows no trend either. These slopes are much smaller than the year-to-year variability. For example, the mean winter (DJFM) H_{m0} varies from 2.60 m (2011) to 4.34 m

(2014), a coefficient of variation of 12 %. The 2013/14 winter was the most energetic in the record, consistent with the exceptional sequence of Atlantic storms observed along the European coast that winter [3]. Low-energy years such as 2010 correspond to strongly negative North Atlantic Oscillation (NAO) winters. This variability is controlled by large-scale atmospheric patterns (NAO, West Europe Pressure Anomaly) [4]. Thirty years of data are therefore too short to separate a possible long-term trend from this natural variability.

Over the 30-year lifetime of the turbine, I do not expect large changes in the *mean* conditions, but significant year-to-year variations must be expected:

1. **Natural variability.** Several stormy or calm winters in a row can change the decadal mean H_{m0} by about 10 %. This matters for fatigue loads and for weather windows during installation and maintenance.
2. **Climate change.** Global wave projections for the North-East Atlantic give small changes in mean H_{m0} by the end of the century (a few percent, mostly a slight decrease), with low agreement between models [5]. Changes in the position of the storm tracks could modify storm intensity and direction more than the mean, so extremes are more uncertain than mean conditions.
3. **Water level.** Sea-level rise (a few tens of cm over 30 years) is negligible compared with 68 m depth, so it will not change the wave regime at the site.

A stationary wave climate is therefore a reasonable assumption for design, as long as margins are added to cover interannual variability and uncertainty in the extremes.

3.2 Question 2 – Most common wave conditions

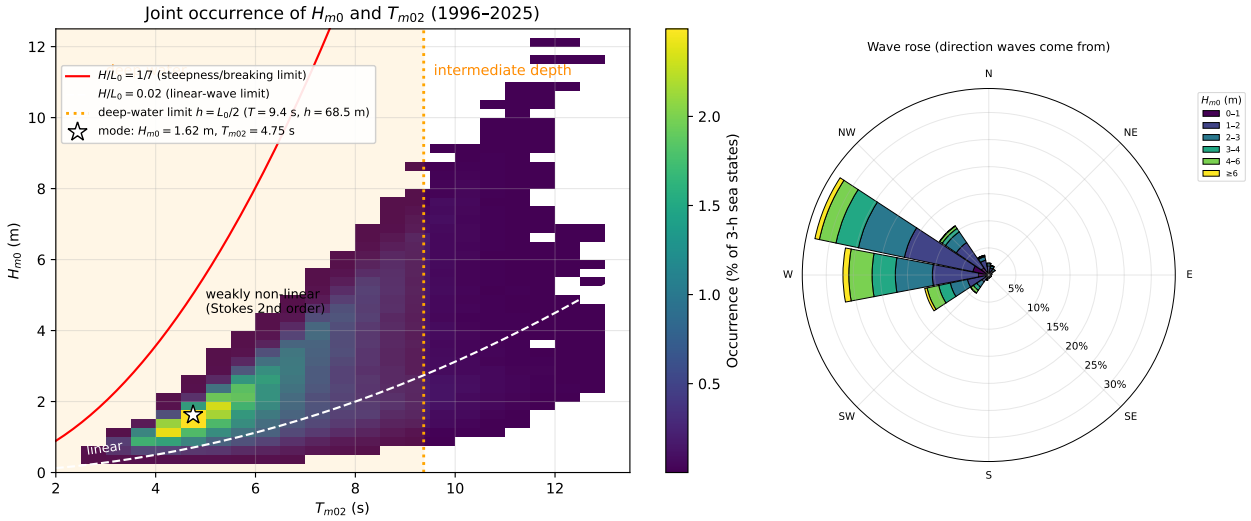


Figure 3: Left: joint occurrence of H_{m0} and T_{m02} (bins of 0.25 m×0.5 s), with the deep-water limit for $h = 68.5$ m (orange; deep water is the shaded side), the linear-wave limit $H_{m0}/L_0 = 0.02$ (white dashed) and the breaking steepness 1/7 (red). Right: wave rose (direction waves come from) coloured by H_{m0}.

The most frequent sea state is $H_{m0} = 1.62$ m, $T_{m02} = 4.75$ s (2184 records, 2.5 % of the time; Fig. 3). More broadly, 36 % of the sea states have H_{m0} between 1 m to 3 m and T_{m02} between 5 s to 7 s. The joint distribution follows a clear steepness relation: high waves are always associated with long periods ($T_{m02} > 9$ s for $H_{m0} > 8$ m).

The wave rose shows a single dominant direction. The four main sectors are WNW (33 %), W (27 %), WSW (12 %) and NW (11 %). In total, 82 % of the waves come from the 225° to 315° sector, and almost all (99 %) of the sea states with $H_{m0} \geq 6$ m come from this sector. This is the open Atlantic fetch: swells generated by North Atlantic depressions propagate freely towards Brittany. Northerly and easterly directions are sheltered by the land (Brittany, Cornwall) and by short fetches.

Q: *What is the water depth at this location? Indicate on the histogram for what wave conditions the waves are considered deep water waves? Linear waves?*

Depth. The mean water depth at the site is $h = 68.5\,\mathrm{m}$ (EMODnet).

Deep water ($h > L_0/2$) corresponds to $T < \sqrt{4\pi h/g} = 9.4\,\mathrm{s}$ (orange shaded area in Fig. 3).

- Using T_{m02}, 95 % of the sea states are deep-water waves.
- The remaining long-period swells and storm waves are in *intermediate depth* and feel the seabed.
- The energetic part of the spectrum is better represented by the peak period T_p, which is about $1.28\,T_{m02}$ for a JONSWAP spectrum. Using T_p, only about 70 % of the sea states are in deep water.
- Shallow-water conditions ($h < L_0/20$, i.e. $T > 30\,\mathrm{s}$) never occur at the site.

Linear waves. The region below the white dashed line ($H_{m0}/L_0 < 0.02$) contains 12 % of the sea states: mainly long, low swells. The remaining 88 % have a steepness between 0.02 and 1/7 (median 0.038). These are *weakly non-linear* waves in the Stokes (second-order) range. No sea state reaches the breaking steepness 1/7. Because the site is deep, non-linearity is controlled by the steepness rather than by the depth:

- $H_{m0}/h \leq 0.18$ even for the largest storm.
- The Ursell number stays small ($U_r \leq 1.7 \ll 26$), so cnoidal or shallow-water theories are not needed.

Linear (Airy) theory is therefore a reasonable first approximation for most conditions. Second-order Stokes corrections should be considered for the steep storm waves used to compute the loads on the structure.

3.3 Question 3 – Seasonal variability

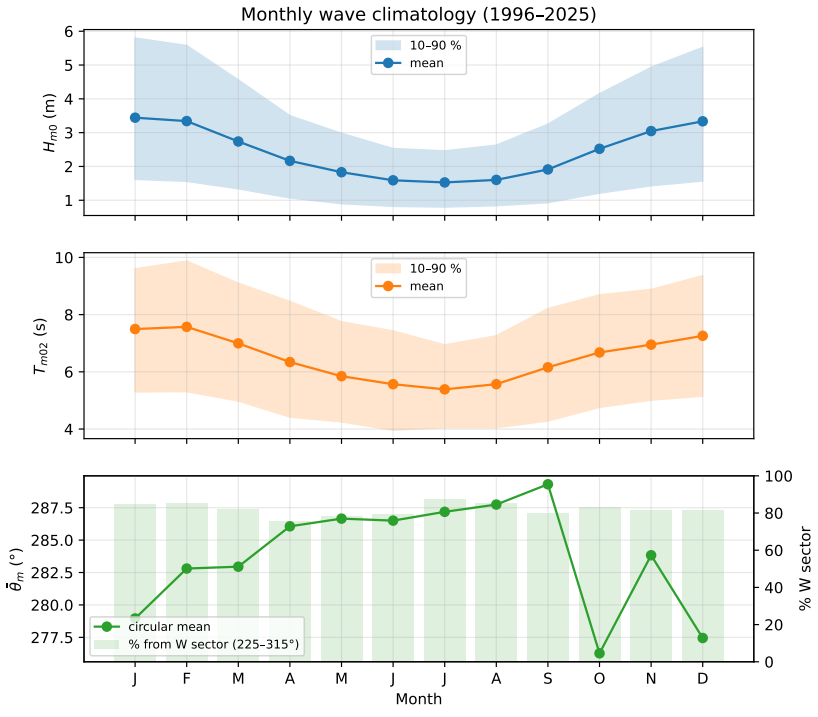


Figure 4: Monthly climatology of H_{m0} and T_{m02} (mean and 10–90 % range) and circular mean direction (line), with the percentage of waves from the W sector, 225° to 315° (bars).

Table 3: Winter (DJF) versus summer (JJA) wave conditions.

	$\overline{H_{m0}}$ (m)	$\overline{T_{m02}}$ (s)	$\bar{\theta}_m$ (°)	Time with $H_{m0} > 4\,\mathrm{m}$
Winter (DJF)	3.37	7.44	280	31 %
Summer (JJA)	1.57	5.51	287	1.1 %

Q: *Do you observe any seasonal trends in wave height, period, or direction? If so, why?*

Yes, there is a strong seasonal cycle in wave height and period, and a weak one in direction (Fig. 4, Table 3).

- **Wave height.** The mean H_{m0} peaks in January (3.44 m) and is lowest in July (1.53 m), so winter waves are about twice as high as summer waves (four times more energetic, since energy scales with H^2). The spread also changes with the season: the 90th percentile reaches 5.8 m in January but only 2.5 m in July. Seas with $H_{m0} > 4\,\mathrm{m}$ occur 31 % of the time in winter and almost never in summer.
- **Period.** T_{m02} follows the same cycle, from 7.6 s in February to 5.4 s in July. Winter seas are dominated by long swells, while summer seas include more short, locally generated wind waves.
- **Direction.** Waves come from the W–WNW all year (80 % to 88 % from the W sector). The mean direction is slightly more westerly in winter (276° to 283°) than in late spring to September (286° to 289°).

Why. The cycle follows the seasonal behaviour of the North Atlantic storm track. In winter, the strong equator–pole temperature gradient produces frequent, deep low-pressure systems. Their strong SW–W winds blow over very long fetches and durations, which generates high, long-period waves and swell travelling towards Brittany. In summer, the Azores high extends north, the storm track moves towards higher latitudes and becomes weaker, and winds are lighter and more often from the NW–N (e.g. sea-breeze or Azores-high flow). As a result, waves are smaller and shorter, with a slightly more north-westerly direction. The higher winter variability (wider 10–90 % range) reflects the alternation between storms and calmer periods. For the project, operation and maintenance should be planned for April–September, and survival loads are governed by winter storms.

4 Discussion

Limitations and uncertainties.

1. **Model data.** The data are model results (MFWAM reanalysis) at 0.2° resolution, about 8 km from the site, without validation against local measurements. Spectral wave models tend to underestimate the peaks of the largest storms. A comparison with nearby buoys (e.g. the CANDHIS network or the Brittany offshore buoys) or with the higher-resolution regional reanalyses (NWSHELF, ResourceCode) is recommended.
2. **Water depth.** Depths from 68 m to 87 m were found within a few km. This changes the deep-water limit ($T < 9.4\,\mathrm{s}$ to 10.6 s) but not the conclusions.

Next steps.

- Download the peak period T_p, which better describes the energetic part of the spectrum than T_{m02}.
- Add the wind and current roses (ERA5, CMEMS physics).
- Estimate the extreme wave conditions (Part I.B).
- Use the typical sea state identified here ($H_{m0} \approx 1.6\,\mathrm{m}$, $T_{m02} \approx 5\,\mathrm{s}$, from 285°) as an operational offshore boundary condition for the wave propagation model (Part II).

5 Conclusion

- **Mean conditions.** The site (68.5 m deep, 9 m/s mean wind) is exposed to the open Atlantic. The mean sea state is $H_{m0} = 2.4\,\mathrm{m}$, $T_{m02} = 6.5\,\mathrm{s}$ from the WNW.

- **Trends.** No significant trend is found over 1996–2025. Interannual variability (about $\pm 12\,\%$ in winter H_{m0}) dominates.
- **Most common conditions.** The typical sea state is $H_{m0} \approx 1.6\,\mathrm{m}$, $T_{m02} \approx 4.75\,\mathrm{s}$. Waves come mainly from the W–WNW and are mostly deep-water, weakly non-linear waves for which linear theory is a reasonable first approximation.
- **Seasonality.** Strong seasonality: winter waves are twice as high and 2 s longer than summer waves.

References

[1] E.U. Copernicus Marine Service Information (CMEMS). *Global Ocean Waves Reanalysis*, product GLOBAL_MULTIYEAR_WAV_001_032. https://doi.org/10.48670/moi-00022.

[2] WindEurope (2017). *Floating offshore wind vision statement.*

[3] G. Masselink et al. (2016). Extreme wave activity during 2013/2014 winter and morphological impacts along the Atlantic coast of Europe. *Geophys. Res. Lett.*, 43, 2135–2143.

[4] B. Castelle et al. (2017). A new climate index controlling winter wave activity along the Atlantic coast of Europe: the West Europe Pressure Anomaly. *Geophys. Res. Lett.*, 44, 1384–1392.

[5] J. Morim et al. (2019). Robustness and uncertainties in global multivariate wind-wave climate projections. *Nature Climate Change*, 9, 711–718.